
Master Thesis

FYSM64

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1 Introduction

[1]

2 Method

3 Result

4 Discussion

5 Lindblad Approach

We write the full Hamiltonian as

$$H = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}} + H_{\text{env}}, \quad (5.1)$$

where H_{det} is the Hamiltonian of the detector defined as

$$H_{\text{det}} = \omega_r a^\dagger a + \frac{\omega_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+), \quad (5.2)$$

H_{sys} is the Hamiltonian of the measured system which is defined, for a DQD in energy basis, as

$$H_{\text{sys}} = \frac{\omega_s}{2} \tau_z, \quad (5.3)$$

H_{int} is the interaction Hamiltonian between the detector and the measured system, defined as

$$H_{\text{int}} = g_d [C \tau_z \sigma_z + S(\tau_+ \sigma_- + \tau_- \sigma_+)], \quad (5.4)$$

and H_{env} is the Hamiltonian consisting of the transmission line, described as

$$H_{\text{env}} = \int d\omega \omega b^\dagger(\omega) b(\omega) + i \sqrt{\frac{\kappa}{2\pi}} \int d\omega [b^\dagger(\omega) a - b(\omega) a^\dagger]. \quad (5.5)$$

From the quantum Langevin equation, we have the following equations of motion

$$\dot{a} = -\left(i\omega_r + \frac{\kappa}{2}\right) a - ig\sigma_- - \sqrt{\kappa_c} b_{\text{in}}(t) \quad (5.6a)$$

$$\dot{\sigma}_- = -(i\omega_q + 2ig_d C \tau_z + \Gamma_2^\sigma) \sigma_- + i(ga + g_d S \tau_-) \sigma_z \quad (5.6b)$$

$$\dot{\tau}_- = -(i\omega_s + 2ig_d C \sigma_z + \Gamma_2^\tau) \tau_- + ig_d S \sigma_- \tau_z \quad (5.6c)$$

$$\dot{\sigma}_z = 2ig(a^\dagger \sigma_- - a \sigma_+) + 2ig_d S(\tau_+ \sigma_- - \tau_- \sigma_+) - \Gamma_1^\sigma(\sigma_z + 1) \quad (5.6d)$$

$$\dot{\tau}_z = 2ig_d S(\sigma_+ \tau_- - \sigma_- \tau_+) - \Gamma_1^\tau(\tau_z + 1), \quad (5.6e)$$

To instead use the Lindblad approach we start by writing the Lindblad master equation for the density matrix as

$$\dot{\rho} = -i[H, \rho] + \sum_i \gamma_i \mathcal{D}[L_i] \rho \quad (5.7)$$

with

$$\mathcal{D}[L_i] \rho = L_i \rho L_i^\dagger - \frac{1}{2} \{L_i^\dagger L_i, \rho\}. \quad (5.8)$$

In this formalism we will have to rewrite the Hamiltonian, since the dissipation to the environment comes from the dissipation terms and to include the drive. We define

$$H_{\text{closed}} = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}}, \quad (5.9)$$

and

$$H = H_{\text{closed}} + H_{\text{drive}}. \quad (5.10)$$

For an expectation value $\langle o \rangle$ the equation of motion can be written as

$$\partial_t \langle o \rangle = \text{tr} \{ o \dot{\rho} \}. \quad (5.11)$$

We require that the equation of motion calculated from the Lindblad equation is the same as the ones calculated from the quantum Langevin equation. Thus, for the operator a we have

$$\partial_t \langle a \rangle = -i \text{tr} \{ a [H, \rho] \} + \sum_i \gamma_i \text{tr} \{ a \mathcal{D}[L_i] \rho \} \quad (5.12a)$$

$$= -i \text{tr} \{ [a, H_{\text{closed}}] \rho \} - i \text{tr} \{ [a, H_{\text{drive}}] \rho \} + \sum_i \gamma_i \text{tr} \{ a \mathcal{D}[L_i] \rho \} \quad (5.12b)$$

by using the cyclic property of the trace. Calculating the commutator for the closed part we find

$$[a, H_{\text{closed}}] = \omega_r a + g \sigma_-. \quad (5.13)$$

It is easy to verify that the only relevant dissipation term is $\gamma_1 = \kappa$ and $L_1 = a$, and

$$\text{tr} \{ a \mathcal{D}[a] \rho \} = -\frac{1}{2} \langle a \rangle. \quad (5.14)$$

To obtain the drive Hamiltonian we take the difference between the two equations of motion and equate

$$-i \text{tr} \{ [a, H_{\text{drive}}] \rho \} = -\sqrt{\kappa_c} \langle b_{\text{in}} \rangle \quad (5.15a)$$

$$0 = \text{tr} \{ i [a, H_{\text{drive}}] \rho - \sqrt{\kappa_c} b_{\text{in}} \rho \} \quad (5.15b)$$

$$0 = i [a, H_{\text{drive}}] - \sqrt{\kappa_c} b_{\text{in}} \quad (5.15c)$$

$$[a, H_{\text{drive}}] = -i \sqrt{\kappa_c} b_{\text{in}}. \quad (5.15d)$$

Naïvely we see that $H_{\text{drive}} = -i \sqrt{\kappa_c} b_{\text{in}} a^\dagger$ solves the equation. However, this cannot be the full Hamiltonian since we require that it to be Hermitian. A simple way to make it Hermitian is to add the Hermitian conjugate, and thus

$$H_{\text{drive}} = -i \sqrt{\kappa_c} (b_{\text{in}} a^\dagger - b_{\text{in}}^\dagger a). \quad (5.16)$$

Carrying out similar calculations for each equation of motion gives the operator and corresponding rates for all Lindblad dissipation terms. The result of the calculations can be seen in Tab. 1.

Putting everything together yields the final Lindblad master equation for this system as

$$\dot{\rho} = -i [H, \rho] + \kappa \mathcal{D}[a] \rho + \Gamma_1^\sigma \mathcal{D}[\sigma_-] \rho + \Gamma_1^\tau \mathcal{D}[\tau_-] \rho + \frac{\Gamma_\phi^\sigma}{2} \mathcal{D}[\sigma_z] \rho + \frac{\Gamma_\phi^\tau}{2} \mathcal{D}[\tau_z] \rho, \quad (5.17)$$

Operator	a	σ_-	τ_-	σ_z	τ_z
Rate	κ	Γ_1^σ	Γ_1^τ	$\Gamma_\phi^\sigma/2$	$\Gamma_\phi^\tau/2$

Table 1: Lindblad jump operators, L_i , and their corresponding rates, γ_i . The possible dissipation channels are photon loss from cavity (a), longitudinal relaxation of the qubits (σ_-, τ_-), and loss of phase coherence (σ_z, τ_z).

with

$$H = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}} + H_{\text{drive}}. \quad (5.18)$$

The next step is to rotate this into the drive frame, such that we remove any time dependence and instead focus on detunings. For a coherent drive we can write

$$b_{\text{in}} = \beta e^{-i\omega_d t}, \quad (5.19)$$

with β real and positive. Defining the unitary operator

$$U(t) = e^{iH_{\text{rot}} t} \quad (5.20)$$

with

$$H_{\text{rot}} = \omega_d \left(a^\dagger a + \frac{\sigma_z}{2} + \frac{\tau_z}{2} \right). \quad (5.21)$$

Rotating the Hamiltonian we get

$$\tilde{H} = U H U^\dagger + i \dot{U} U^\dagger \quad (5.22)$$

and introducing

$$\Delta_r = \omega_r - \omega_d \quad (5.23a)$$

$$\Delta_q = \omega_q - \omega_d \quad (5.23b)$$

$$\Delta_s = \omega_s - \omega_d \quad (5.23c)$$

we can write the full rotated Hamiltonian, dropping the tilde for simplicity, as

$$H = \overbrace{\Delta_r a^\dagger a + \frac{\Delta_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+)}^{\text{Detector}} + \overbrace{\frac{\Delta_s}{2} \tau_z + g_d [C \tau_z \sigma_z + S(\tau_+ \sigma_- + \tau_- \sigma_+)]}_{\text{System-detector interaction}} + \underbrace{i \sqrt{\kappa_c} \beta (a - a^\dagger)}_{\text{Drive}} \quad (5.24)$$

5.1 Some Approximations

To obtain an analytical expression to compare with numerical solutions we require some approximations. Since we want the measurement to be QND, and we can, by construction of the qubits, chose the tunnelling rates such that the mixing angles make $S = 0$. We will also say that, since phase decoherence generally dominate we take the relaxation to be $\Gamma_1^\sigma = \Gamma_1^\tau = 0$. We will also consider a mean field interaction. Using these approximations τ_z becomes a constant of motion and the reflection coefficient can be written as

$$r = 1 - \frac{\kappa_c}{\frac{\kappa}{2} + i\Delta_r + \frac{g^2}{\Gamma_\phi^\sigma + i(\Delta_q + 2g_d C \tau_z)}}, \quad (5.25)$$

with $\tau_z = \pm 1$.

5.2 Measurement rate

From Clerk *et al.* [2] we know that the measurement rate is bound by the measurement induce dephasing rate, $\Gamma_{\phi,\text{meas}}^\tau$. Furthermore, in the long time limit, we can write this measurement induced dephasing rate as

$$\Gamma_{\phi,\text{meas}}^\tau = \frac{\kappa}{2} \left| \langle a \rangle_e^{\text{ss}} - \langle a \rangle_g^{\text{ss}} \right|^2, \quad (5.26)$$

where the subscript e/g denotes if the external qubit is in the excited or ground state, while ss denotes that it is the steady state [3]. Thus, we have an upper bound on the measurement.

6 Calculation of the Quantum Limit

We write the full Hamiltonian as

$$H = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}} + H_{\text{env}}, \quad (6.1)$$

where H_{det} is the Hamiltonian of the detector defined as

$$H_{\text{det}} = \omega_r a^\dagger a + \frac{\omega_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+), \quad (6.2)$$

H_{sys} is the Hamiltonian of the measured system which is defined, for a DQD in energy basis, as

$$H_{\text{sys}} = \frac{\omega_s}{2} \tau_z, \quad (6.3)$$

H_{int} is the interaction Hamiltonian between the detector and the measured system, defined as

$$H_{\text{int}} = g_d [C \tau_z \sigma_z + S(\tau_+ \sigma_- + \tau_- \sigma_+)], \quad (6.4)$$

and H_{env} is the Hamiltonian consisting of the environment causing relaxation and dephasing, described as

$$H_{\text{env}} = H_{\text{cavity}} + H_{\text{rel}}^\sigma + H_\phi^\sigma + H_{\text{rel}}^\tau + H_\phi^\tau. \quad (6.5)$$

The cavity Hamiltonian is defined as

$$H_{\text{cavity}} = \int d\omega \omega b^\dagger(\omega) b(\omega) + i \sqrt{\frac{\kappa}{2\pi}} \int d\omega [b^\dagger(\omega) a - b(\omega) a^\dagger], \quad (6.6)$$

the relaxation Hamiltonians for the qubits are defined similarly but with Γ_1^σ and Γ_1^τ instead of κ and their own bosonic operators, $\xi_{\phi/\sigma}$. The dephasing Hamiltonians are defined by

$$H_\phi^\sigma = \int d\omega \omega \xi_\phi^\dagger(\omega) \xi_\phi(\omega) + \sigma_z \int d\omega \lambda_\omega [\xi_\phi^\dagger(\omega) + \xi_\phi(\omega)] \quad (6.7)$$

We define

$$H_{\text{closed}} = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}}. \quad (6.8)$$

Using Heisenberg's equation of motion we obtain

$$\dot{b}(\omega) = -i[b(\omega), H] = -i[b(\omega), H_{\text{env}}] = -i\omega b(\omega) + \sqrt{\frac{\kappa}{2\pi}} a \quad (6.9)$$

with solution

$$b(\omega, t) = e^{-i\omega(t-t_0)}b(\omega, t_0) + \sqrt{\frac{\kappa}{2\pi}} \int_{t_0}^t dt' e^{-i\omega(t-t')} a(t'). \quad (6.10)$$

We also define the quantum Langevin equation for an operator o as

$$\dot{o} = -i[o, H] + \sum_l \left[-[o, a_l^\dagger] \left(\frac{\kappa_l}{2} a_l + \sqrt{\kappa} b_{\text{in},l}(t) \right) + \left(\frac{\kappa}{2} a_l^\dagger + \sqrt{\kappa_l} b_{\text{in},l}^\dagger(t) \right) [o, a_l] \right], \quad (6.11)$$

where the H is the Hamiltonian minus the bath and $l \in \{c, i\}$ for c being the coupling port to the transmission line and i being the internal losses. It is also noted that $a_c = a_i = a$, $b_{\text{in},c} = b_{\text{in}}$, $b_{\text{in},i} = \xi$, and $\kappa = \kappa_c + \kappa_i$. Applying the quantum Langevin equation to the system we get the following equations of motion

$$\dot{a} = -\left(i\omega_r + \frac{\kappa}{2}\right)a - ig\sigma_- - \sqrt{\kappa_c}b_{\text{in}}(t) - \sqrt{\kappa_i}\xi(t) \quad (6.12a)$$

$$\dot{\sigma}_- = -(i\omega_q + 2ig_d C\tau_z + \Gamma_2^\sigma)\sigma_- + i(ga + g_d S\tau_-)\sigma_z \quad (6.12b)$$

$$\dot{\tau}_- = -(i\omega_s + 2ig_d C\sigma_z + \Gamma_2^\tau)\tau_- + ig_d S\sigma_- \tau_z \quad (6.12c)$$

$$\dot{\sigma}_z = 2ig(a^\dagger\sigma_- - a\sigma_+) + 2ig_d S(\tau_+\sigma_- - \tau_-\sigma_+) - \Gamma_1^\sigma(\sigma_z + 1) \quad (6.12d)$$

$$\dot{\tau}_z = 2ig_d S(\sigma_+\tau_- - \sigma_-\tau_+) - \Gamma_1^\tau(\tau_z + 1), \quad (6.12e)$$

where we have

$$\Gamma_2 = \frac{\Gamma_1}{2} + \Gamma_\phi. \quad (6.13)$$

References

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